

A Term or not a Term? – That is the Question

ARC Terms Mark-up Taskforce Briefing

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Faculty of Applied Sciences, Ukrainian Catholic University, Lviv – Ukraine

Fill out this form to join the taskforce: <https://forms.gle/e6VnNpGHPjLtaEcw8>

Myself ... Following an Italian Colleague ...

2 Teaching ~ 50% of time

- **Prof. of Semantic Technologies**, UCU, AppS, Dept of CS & IT; PhD Advisor; Teach some courses (B.Sci., M.Sci., Ph.D.)

2 Administration ~ 50% of time

- **Director**, Ph.D. Program in Intelligent Systems, UCU, AppS

1 Research ~ 50% of time

- AI: LLM, NLP, Knowledge Engineering, Semantic Technologies
- A bunch of projects (FP6, FP7, BMBF, Interreg, ETF, National): **Researcher**, team lead, etc.;
- **Research consultant** for public organizations and industries

Order matters

- + **Community Service** – also some time
- **Founder**: ICTERI conference series;
External expert: EC FP6, 7, H2020, HE LangTech & Digital,
PC member, **EB member**



More a Research Professor

The SIFR-2026 Team



- Victoria Kosa, PhD
- SIFR-2026 coordinator



- Yuliana Hrynda
- Terms mark-up taskforce coordinator

- Please contact Victoria
 - victoriya.kosa@ucu.edu.ua
 - In any matter related to SIFR-2026 administration

- Please contact Yuliana
 - hrynda.pn@ucu.edu.ua
 - In any matter related to your individual work on terms mark-up

Plan

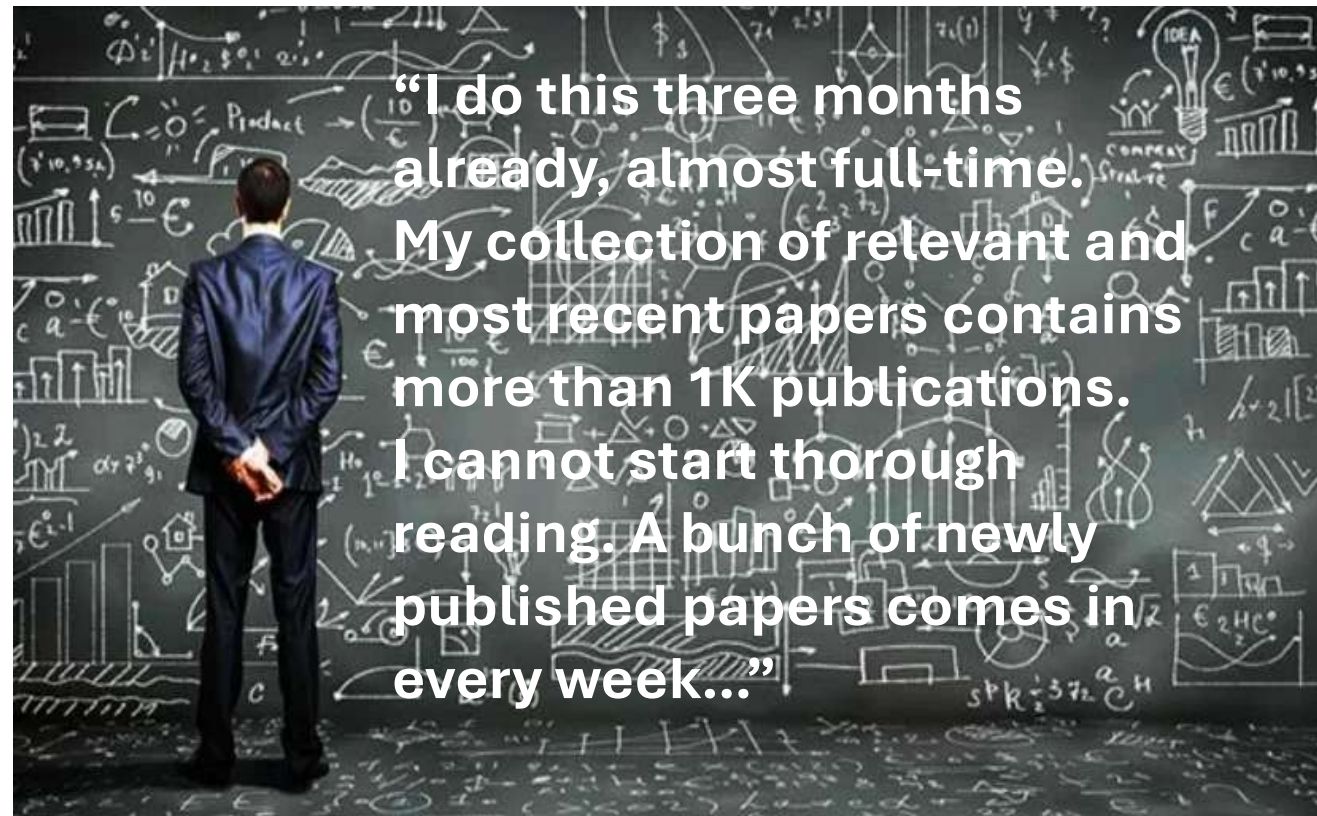
- **Motivation:** Scholarly domains understanding
- **Research Frontier:** Building KGs from representative text corpora
- **Problem:** A term or not a term? – Recognise and Categorise T-s
 - Definition(s)
 - Recognition: Why difficult ...
 - Evaluation
- **Task:** Build a Golden Dataset for a Benchmark
 - Domain
 - Data (papers)
 - Method
- **Organization**

Motivation: Knowledge Acquisition Bottleneck

New information comes in quicker than processed into knowledge

E.g.:

- A 1st-year Ph.D. student
- Working on SotA review
- Any scholarly domain/field
- Typical (not only) for Ph.D. students
- **NEEDED: Instruments to pass through the KAB**
 - Also AI-enabled



<https://t.ly/jPKdH>

Motivation: Rabbit's Hole

AI-Enabled ...

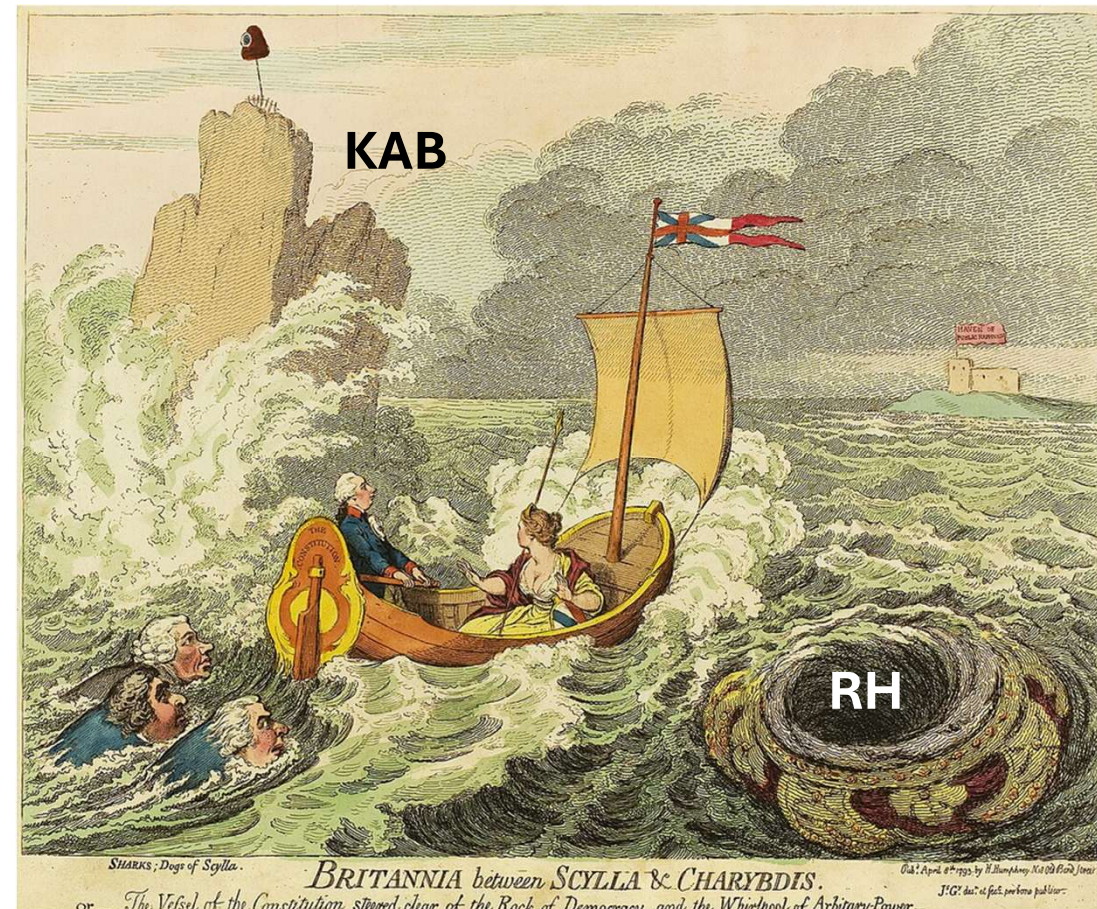
- I used ChatGPT for:
 - Summarizing the discovered papers
 - Recommending the most relevant/valuable papers
- I found myself in the Rabbit's Hole:
 - Awakening to a new way of seeing things
 - Not sure, though, that this alternative maps into reality
 - Reason: lack of explainability
- Verification – extra effort



Question: How to Pass Between ...

... the Scylla and Charybdis
in Scholarly Domains
Understanding?

- Our **Answer:**
 - Research & develop an iterative Neuro-Symbolic approach
 - To construct a Knowledge Graph for your domain
 - With you (a researcher) in the loop



https://en.wikipedia.org/wiki/Between_Scylla_and_Charybdis

A dream ...

Shantanu Ladhwe: 9 papers you should know that shaped modern AI.
LinkedIn. Oct. 2025 <https://surl.li/dgwztu>

Existing
AI papers

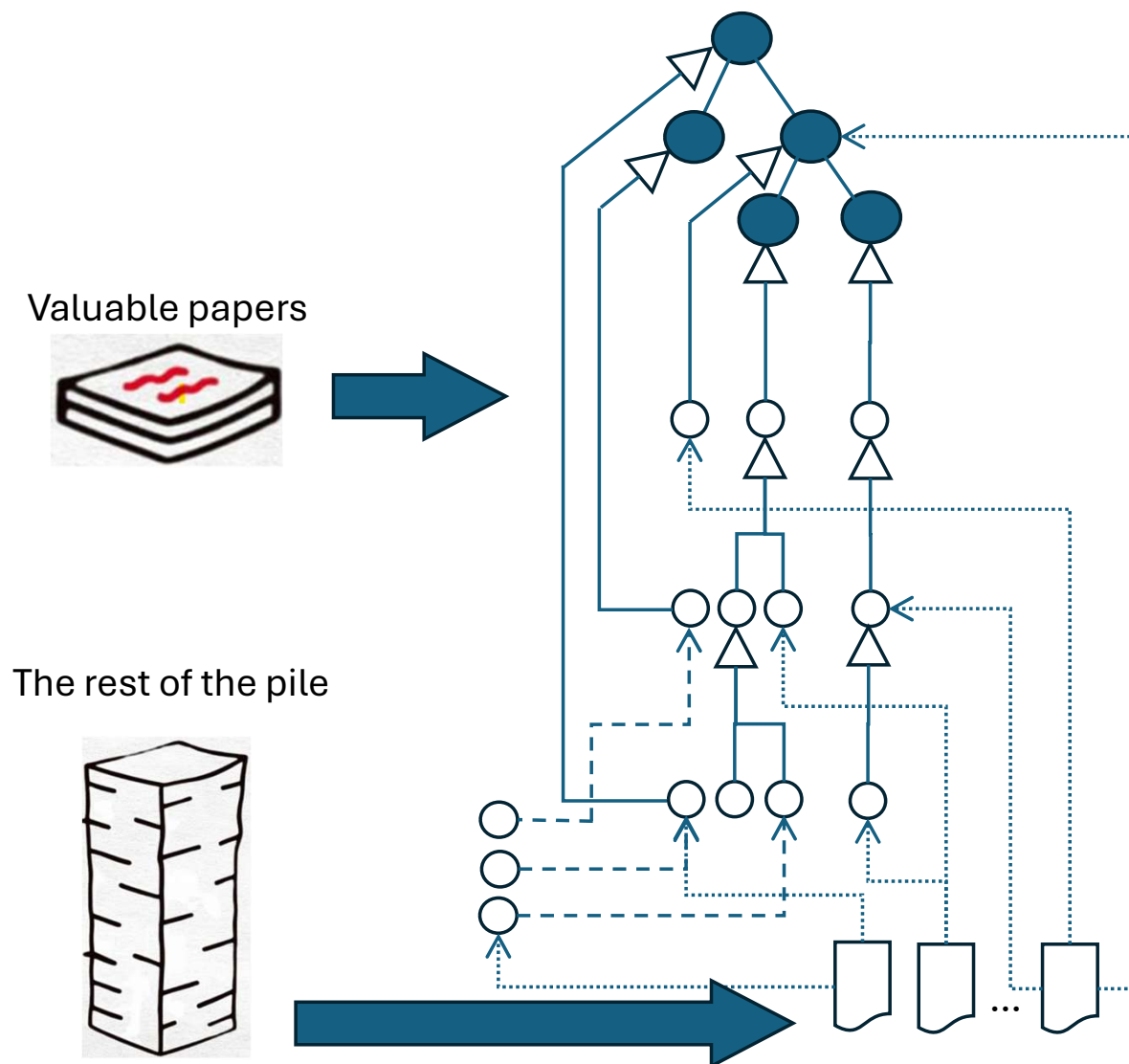
Raw
corpus of
relevant
papers
(yet to be
collected)



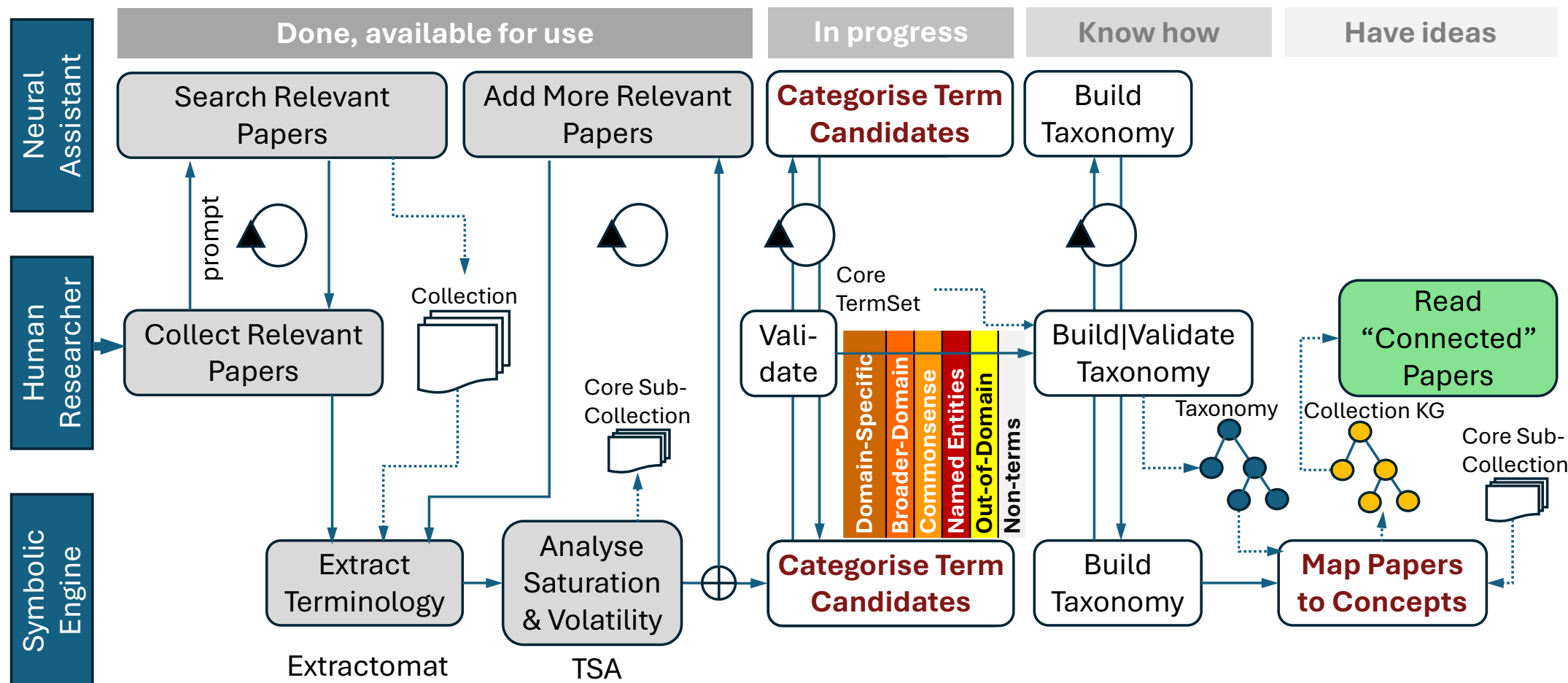
Repre-
sentative
Core sub-
collection
(yet to be
selected)

Valuable
AI papers



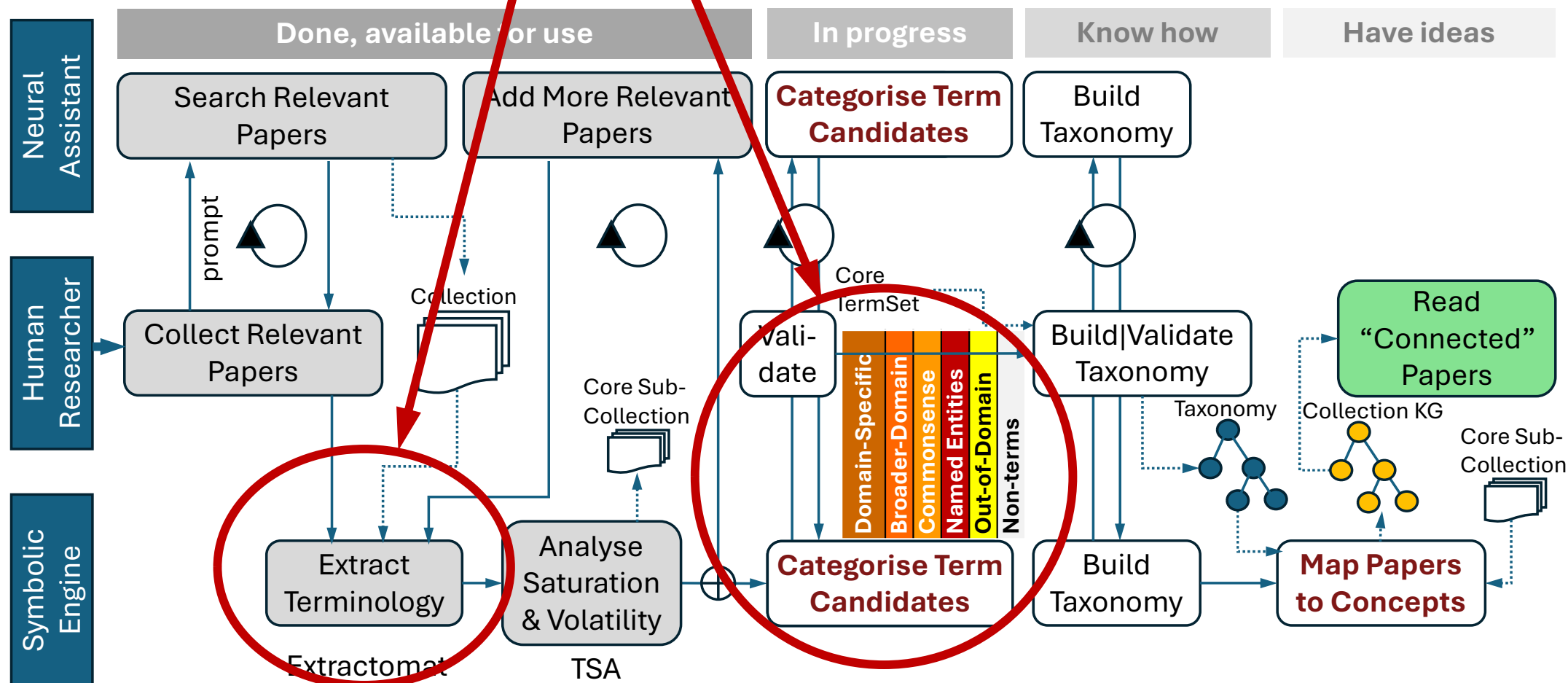


Research in Progress



Our Foci

Evaluate



To Be Done:

NOW

Build a **golden dataset** for evaluating terms recognition and categorization:

- Scholarly domain: Abstraction and Reasoning Challenge (ARC)
- Language: English



NEXT TIME

Extend to:

- Domains: +Anti-Corruption, +more ...
- Languages: +UA, +DE, +FR (parallel corpora)

Are Recognized Candidates Indeed Terms?

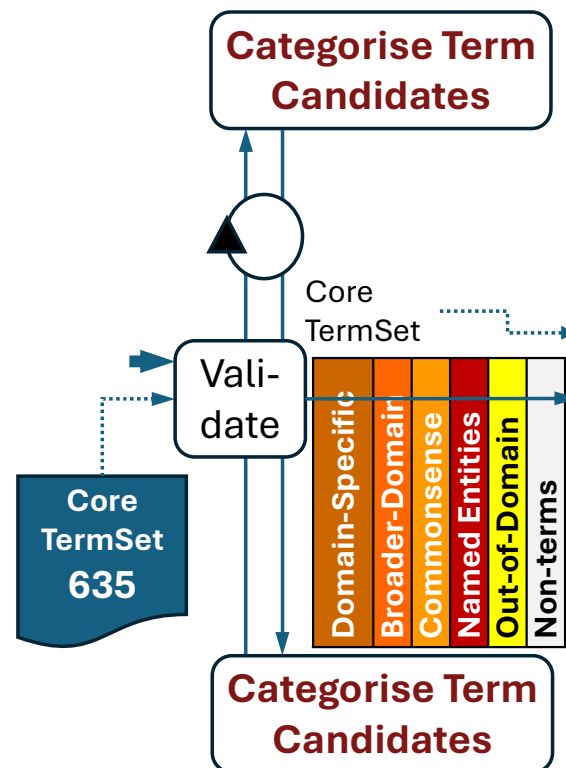
- Recognition
- Categorisation

task	344
intelligence	250
system	248
skill	225
generalization	170
test	151
program	144
t	144
prior	117
evaluation	115
ability	113
human	99
skill program	86.76
knowledge	86
intelligent system	85.59
situation	84
training	75
set	70
?	67
generalization	
difficulty	62.55
...	

Neural Assistant

Human Researcher

Symbolic Engine



What is a Term?



“... a **word** or **expression**
that has a precise **meaning**
in some **uses** or is peculiar to
a **science, art, profession,**
or **subject**”



Merriam-Webster Dict: <https://www.merriam-webster.com/dictionary/term>

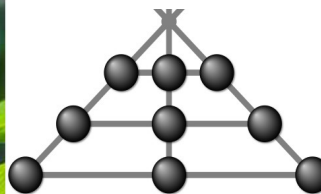
Automatically Recognising General Terms is Hard

- Filtering out the irrelevant candidates
 - Even harder
- E.g.: Is “butterfly” a valid term for AI?

- **Def: butterfly?**
 - Game
 - Swimming technique
 - Goalkeeping technique
 - Movie (any of a ~dozen)
 - ...
 - Yet, not mentioning the obvious



<https://en.wikipedia.org/wiki/Butterfly>



[https://en.wikipedia.org/wiki/Butterfly_\(game\)](https://en.wikipedia.org/wiki/Butterfly_(game))

https://en.wikipedia.org/wiki/Butterfly_style



[wikipedia.org/wiki/Butterfly_stroke](https://en.wikipedia.org/wiki/Butterfly_stroke)



Automatically Recognising General Terms is Hard

- Filtering out the irrelevant candidates

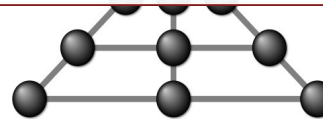
- Even harder: “In **AI**, the **butterfly effect** refers to how a small initial change in a complex system can lead to large, unpredictable outcomes, impacting everything from AI fairness and bias to business transformations.”
- E.g.: Is “butterfly” a valid term for CS?
- Def: butterfly?

- Game of Thrones
- Swimming techniques
- Goalkeeper
- Film (any of a ~dozen)
- NE butterfly

- Yet, not mentioning the obvious



<https://en.wikipedia.org/wiki/Butterfly>



[https://en.wikipedia.org/wiki/Butterfly_\(game\)](https://en.wikipedia.org/wiki/Butterfly_(game))

https://en.wikipedia.org/wiki/Butterfly_style



https://en.wikipedia.org/wiki/Butterfly_stroke



More Complications ...

In addition to pragmatic contexts, polysemy, ambiguity, etc.

- Are these expressions terms?
 - “**Golden Gate Bridge**” in Infrastructure
 - “**Donald Trump**” in State Governance
 - “**Tequila Sunrise**” in Movies, Sportswear, Cocktails, Music, ...
 - “**The Butterfly Effect**” in Movies
- So, are **Named Entities** – terms?
 - Label
 - Instances of ...



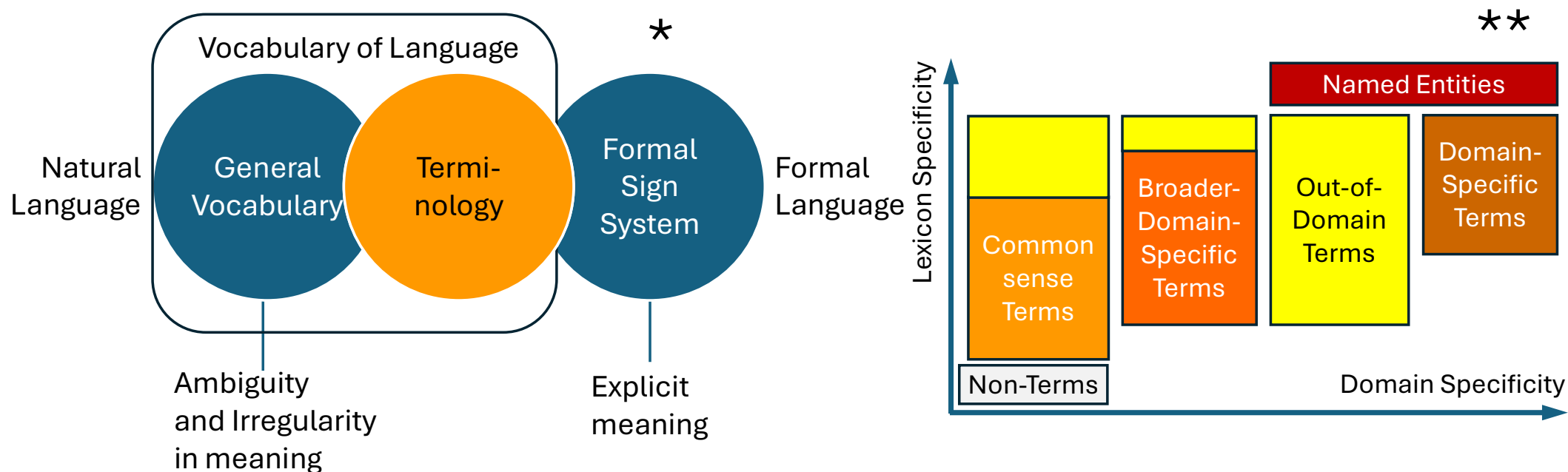
https://en.wikipedia.org/wiki/Golden_Gate_Bridge

... a **word** or **expression** that has a precise meaning in some uses or is peculiar to a **science, art, profession, or subject**

Where is a Term in a Language?*



Is “butterfly effect” a term? Which kind of? Is “The Butterfly Effect” a term?

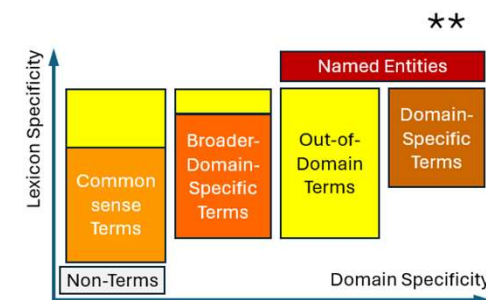


* Kageura K. The quantitative analysis of the dynamics and structure of terminologies. Terminology and lexicography research and practice. Amsterdam: John Benjamins Publishing Company (2012)

** Ermolayev, V.: Passing between a KAB and Rabbit's Hole in Scholarly Domains Understanding. Invited talk. TIB, Leibniz University of Hannover, Dec. 2025 DOI: <https://doi.org/10.13140/RG.2.2.13762.75203>

What is a Term? – Definitions

- 5 outdated, though still frequently cited:



Sager (1990): “A **surface representation** of a **specific-domain concept**.”

Wüster (1979): “A **concept reference system** within a **specialized field**.”

ISO 1087-1 (2000/2008): “A verbal designation of a **general concept** in a **specific field**.”

Kageura & Umino (1996): Term = **word or phrase** “related to **domain-specific concepts**” (encompasses termhood)

Justeson & Katz (1995): Terms are “**domain-specific noun phrases** identified by POS patterns and statistical scores.”

- One more recent:

San Martín, A. (2022)*: “... terms **do not carry meaning in themselves**. They are **access points** to large repositories of knowledge, i.e., their **semantic potential** ...”

* San Martín, A. (2022). Contextual Constraints in Terminological Definitions. *Frontiers in Communication*, 7. <https://doi.org/10.3389/fcomm.2022.885283>

Terms: Categories and Recognition

- Current SotA
- **Challenge**
 - Tacit and **blurred** difference between a term and non-term
 - Also, between different categories of terms

• **Way to attack** the challenge

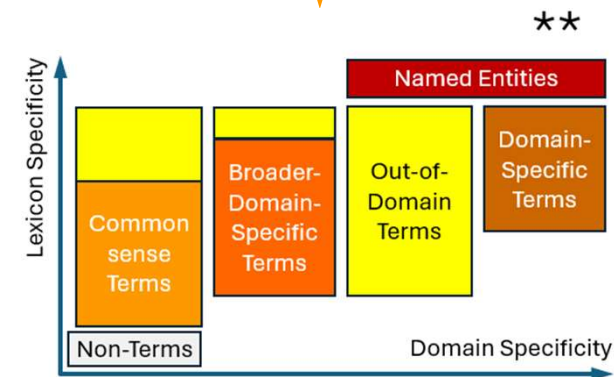
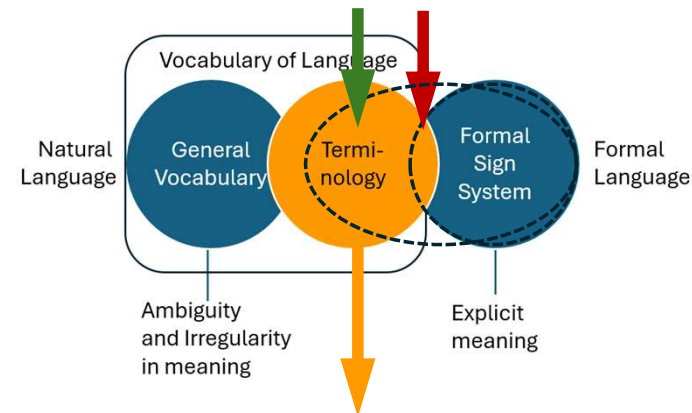
- Expand & refine the Formal Sign System

• **Approach:**

- Reduce the **impedance mismatch**
 - Between tacit lexical; and
 - Explicit semantic representations
- Build a **Lexical-Semantic Bridge**:
 - E.g. Text → WordNet → Ontology → RAG → Knowledge Graph

San Martín, A. (2022):
“... terms **do not carry meaning in themselves**. They are **access points** to large repositories of knowledge, i.e., their **semantic potential ...**”

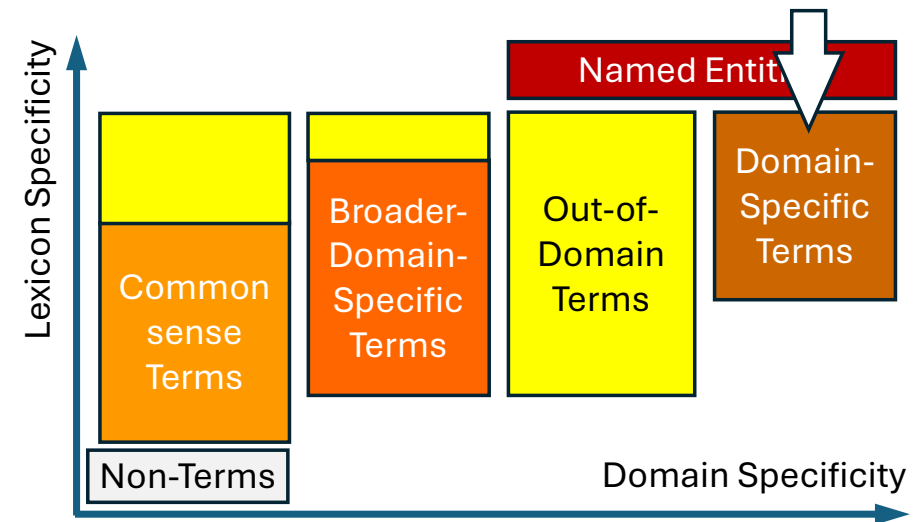
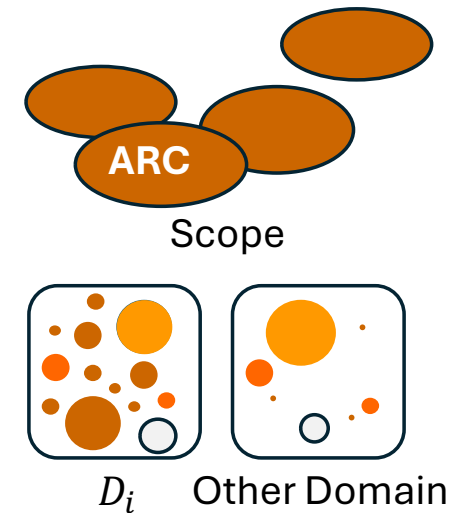
... a **word** or **expression** that has a precise **meaning** in some **uses** or is peculiar to a **science, art, profession, or subject**



Refined Term Categories

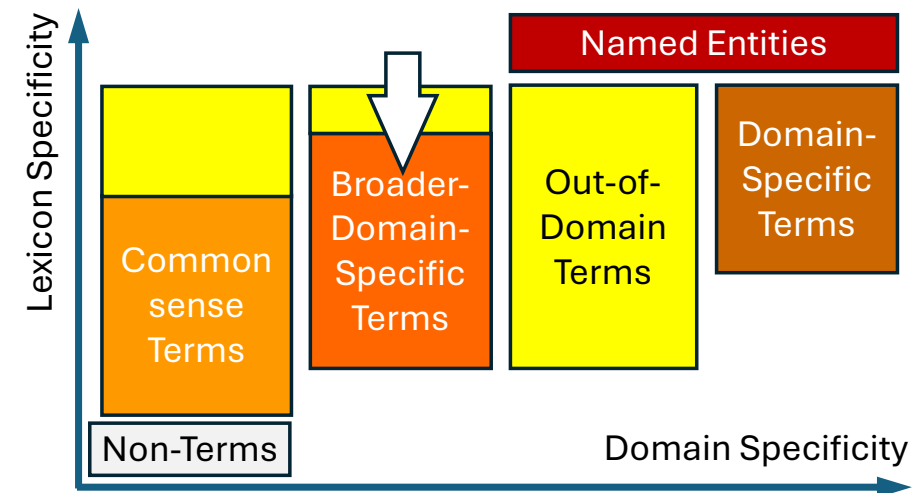
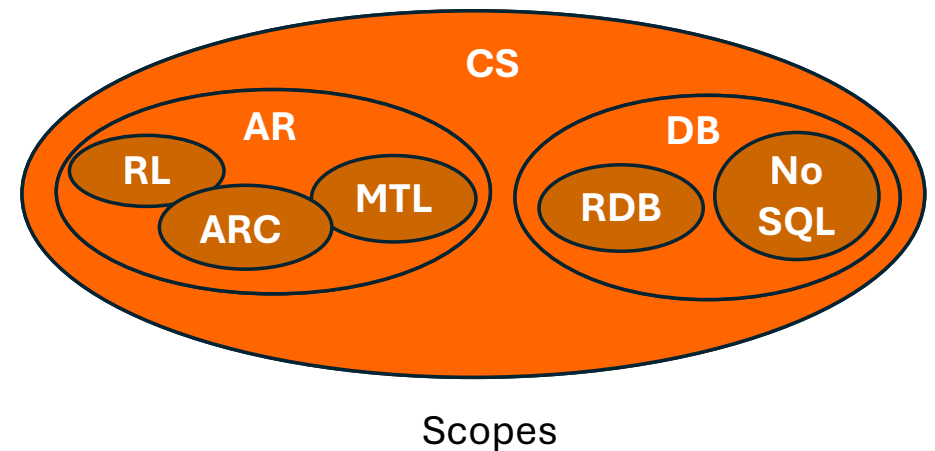
DST – Scope: focal domain D_i

- Have singular, consensual, and well-shaped meanings within D_i
- Are **access points** to conceptualisations in D_i
 - E.g., Through **definitions**
 - E.g.: In ARC domain
- Could occur in the other domains
 - Either rarely – domains overlap
 - Or with a different meaning – another term



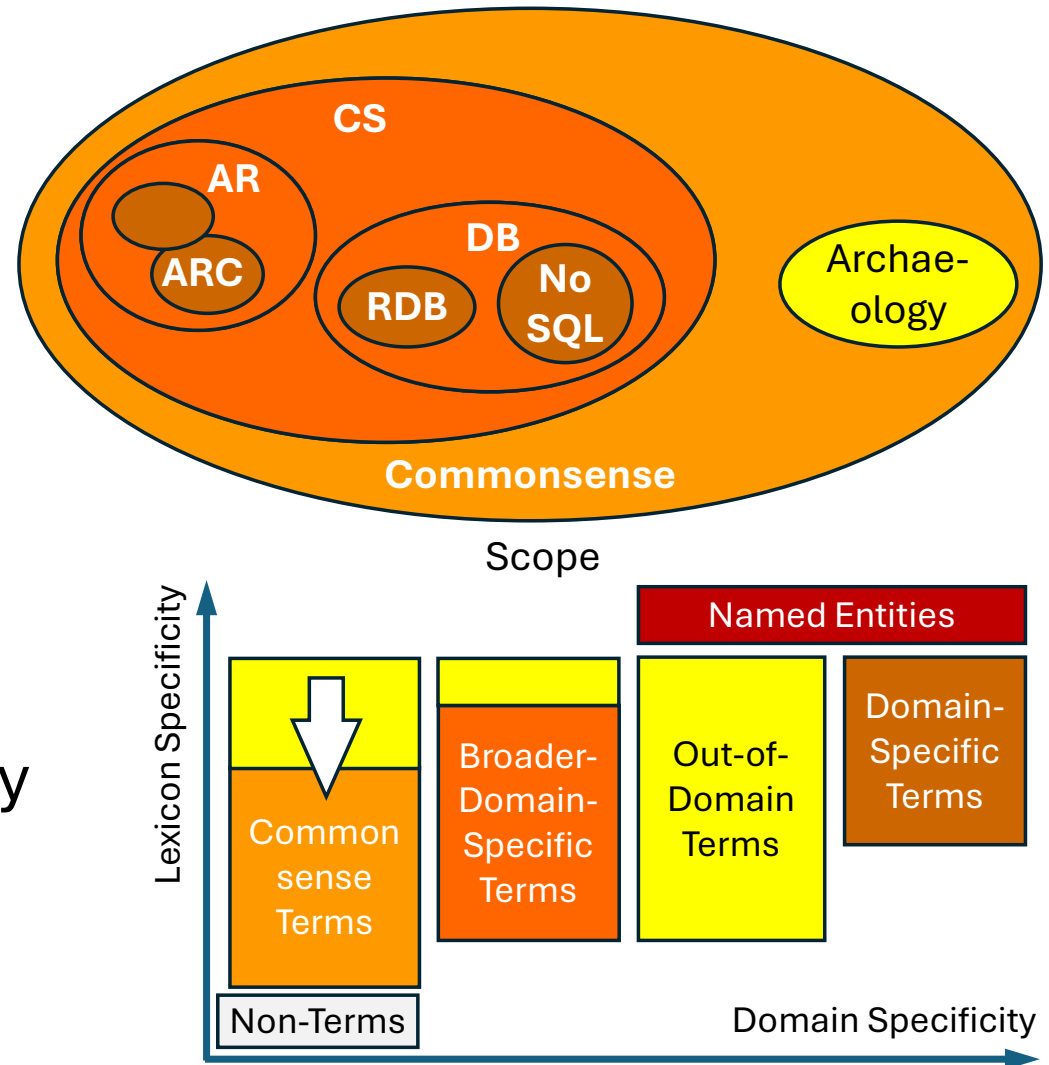
Refined Term Categories

- BDST** – Scope: broader domain covering the basics of several more focused sub-domains
- E.g.: AR for ARC, RL, MTL ...
 - Also: CS for AI for AR, DB, NLP, ...
 - Broader domains form hierarchies
 - BDSTs point to the **middle layers** of the corresponding concept hierarchies
 - DST (concepts) of the covered domains
 - Are often subsumed by BDST
 - Are denoted using these BDST (and CST)



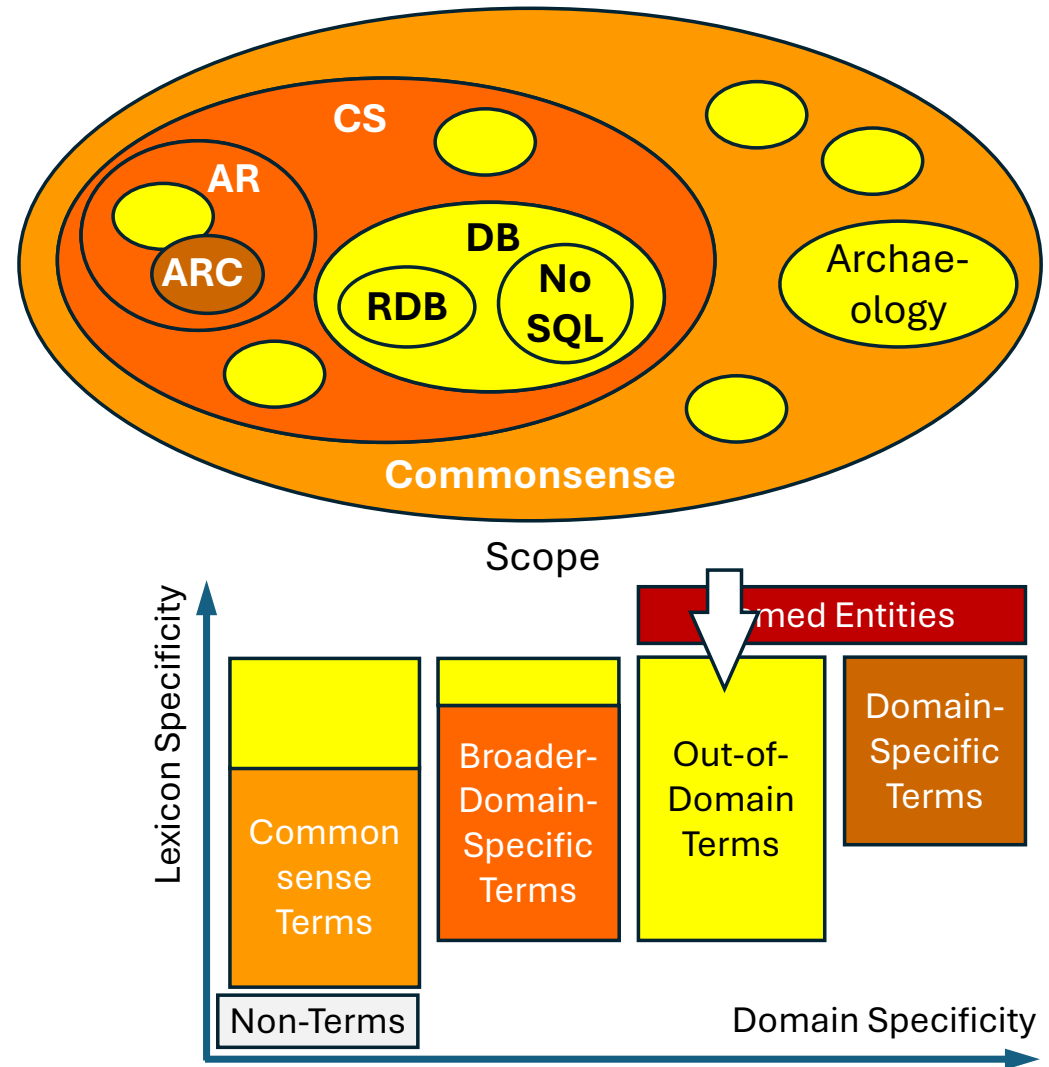
Refined Term Categories

- CST** – Scope: most abstract and foundational terms
- In all existing domains
 - Including irrelevant for D_i (of DST)
 - E.g.: In Archaeology
 - Point to the concepts of foundational ontologies
 - **Top level** of a concept hierarchy
 - Most abstract BDST and DST (concepts) are subsumed by CST concepts



Refined Term Categories

- ODT** – Scope: terms in a different non-overlapping domain
- E.g.: Archaeology
 - Also include:
 - BDST that do not subsume DST from D_i
 - CST that do not (indirectly) subsume DST from D_i



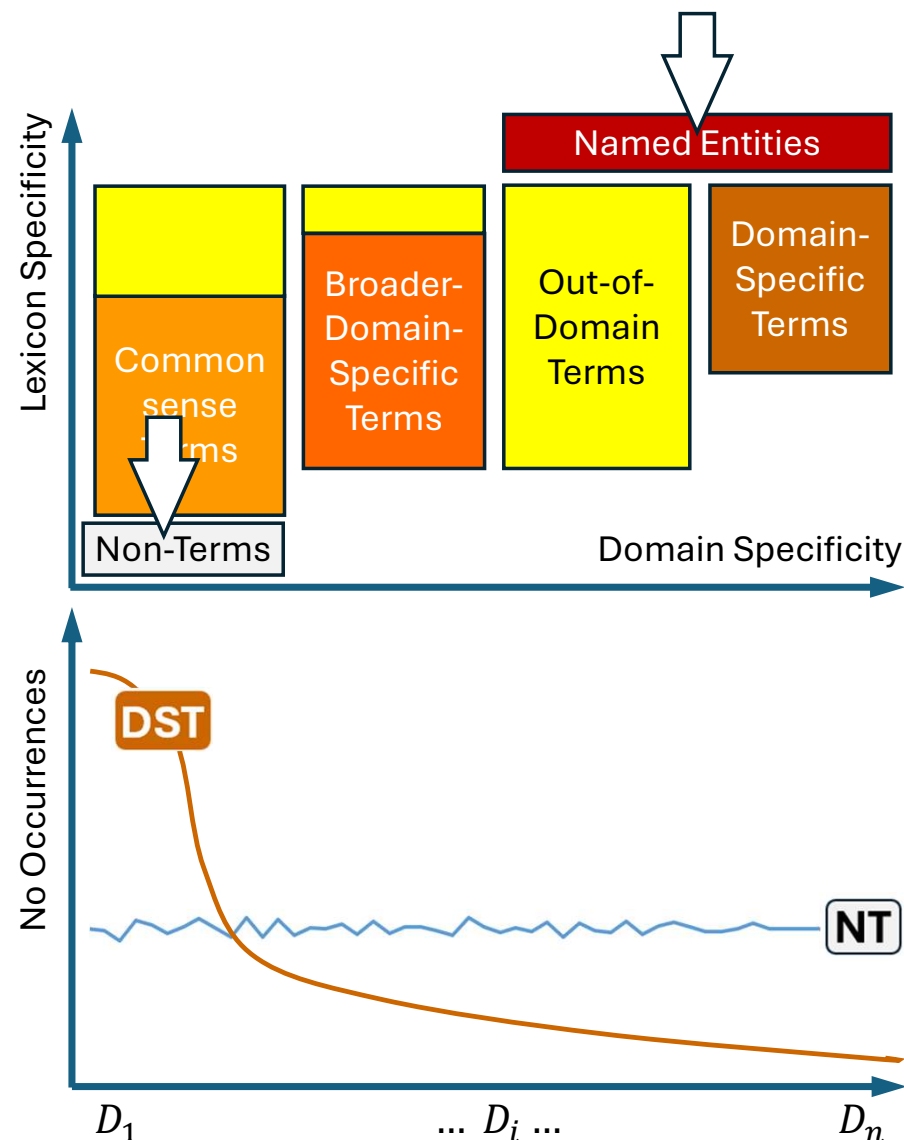
Refined Term Categories

NE – just named entities

- Label **concept instances**
- To be categorised using an available SotA neural NER solution

NT – n-grams that:

- May correspond to PoS patterns
 - E.g., are noun phrases: “butterfly effect”
- Are not terms
 - Are mistakenly selected as term candidates
 - Are regular noise / buzzwords ...
 - If noise, are distributed ~uniformly in the corpus / dataset / collection



The Workflow

- Collect ARC papers
 - Check relevance
- Check if the collected corpus is representative
 - Check terminological saturation
- Retain the Terminological Core Sub-collection
 - Mainstream (saturation) and innovative (volatility)

DONE
By Us

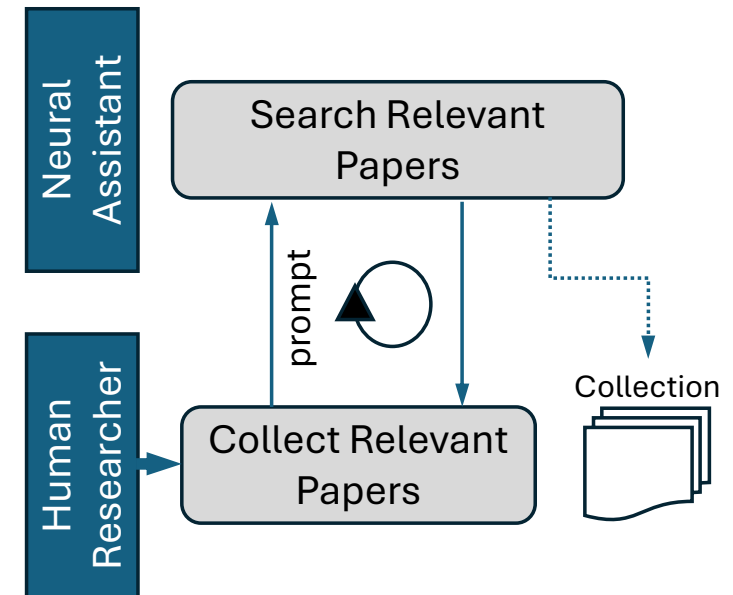
- Manually recognise and categorise the terms in the papers

TBD
by You

Domain: Abstraction and Reasoning Challenge

Collecting Papers 1/5

- One of our PhD students
 - Collected the initial set of pubs
 - Manually: ~50 papers
 - Sources: Connected Papers, arXiv, Google Scholar, Kaggle, etc.
 - Prompted ChatGPT 4o: relevance check



Prompt:

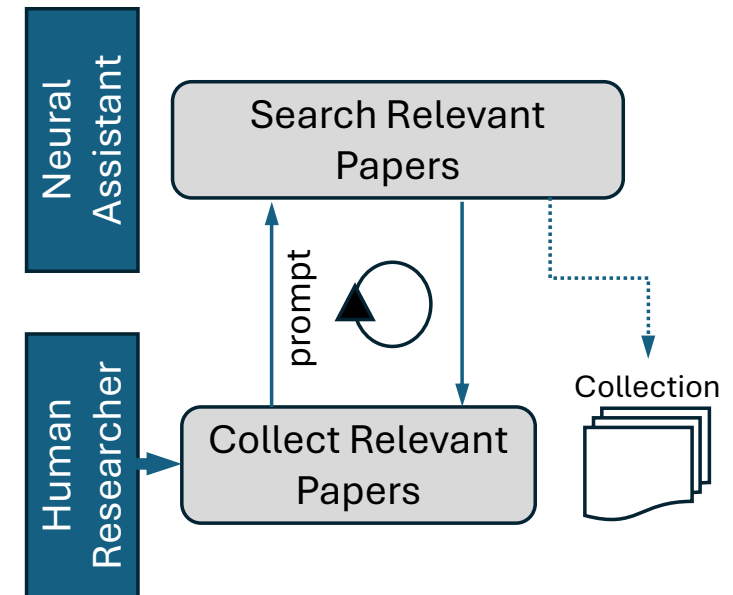
I have the list of publication references:
<list of refs to collected papers>
Please check the list and find out to what topic these
pubs belong.

Reply: The list covers research at the intersection of **cognitive science, program synthesis, meta-learning, and reinforcement learning**, with a **strong emphasis on the Abstraction and Reasoning Corpus (ARC)** and the broader ARC Challenge.

Domain: Abstraction and Reasoning Challenge

Collecting Papers 2/5

- One of our PhD students
 - Collected the initial set of pubs
 - Manually: ~50 papers
 - Sources: Connected Papers, arXiv, Google Scholar, Kaggle, etc.
 - Prompted ChatGPT 4o: irrelevance check



Prompt:

I have the list of publication references:
<list of refs to collected papers>
Please check the list and find the pubs irrelevant for the identified topic.

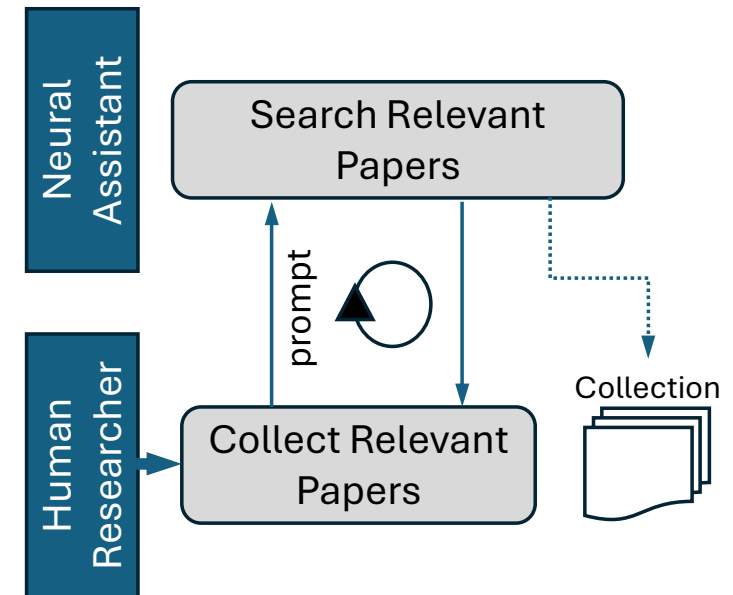
Reply:

None of the listed publications are irrelevant for the topic of **Abstraction and Reasoning Corpus (ARC)** and the broader ARC Challenge.

Domain: Abstraction and Reasoning Challenge

Collecting Papers 3/5

- One of our PhD students
 - Collected the initial set of pubs
 - Manually: ~50 papers
 - Sources: Connected Papers, arXiv, Google Scholar, Kaggle, etc.
- Prompted ChatGPT 4o – completeness



Prompt:

Please suggest additional publications to complete this collection. 50 additional publications may suffice.

Reply: Suggested 50 pubs

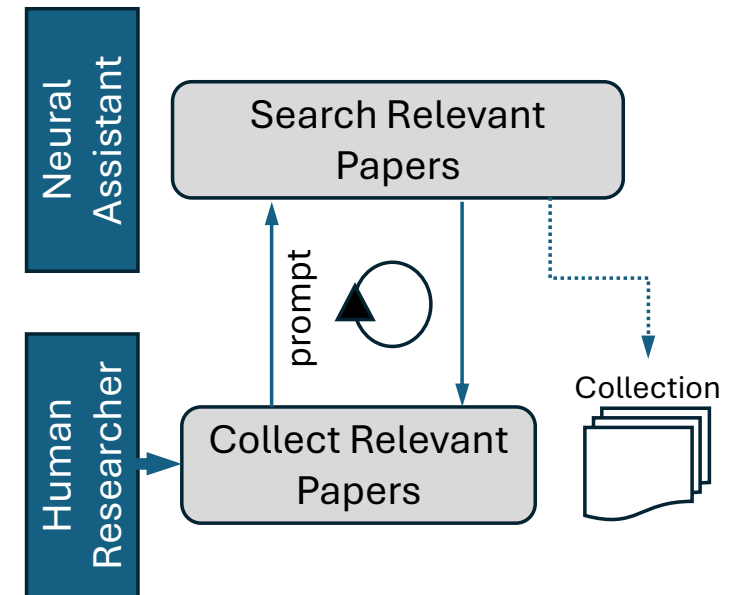
Around 30 of these were
- Either in the initial collection
- Or **not fully relevant**.
~20 pubs added to the collection



Domain: Abstraction and Reasoning Challenge

Collecting Papers 4/5

- One of our PhD students
 - Collected the initial set of pubs – manually:
~50 papers
 - Sources: Connected Papers, arXiv, Google Scholar, Kaggle, etc.
- Queried SCOPUS: ~60 more papers



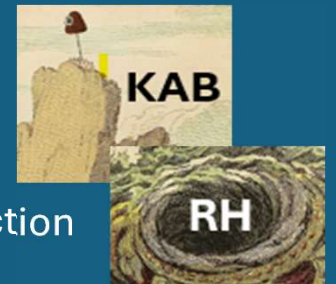
Query:

(ALL ("abstraction and reasoning corpus") AND
PUBYEAR > 2019 AND PUBEYEAR < 2026) AND (arc)
AND PUBEYEAR > 2019 AND PUBEYEAR < 2026 AND (
LIMIT-TO (LANGUAGE , "English"))

Reply: Suggested 60 papers

Around 20% of these were
in the initial collection.

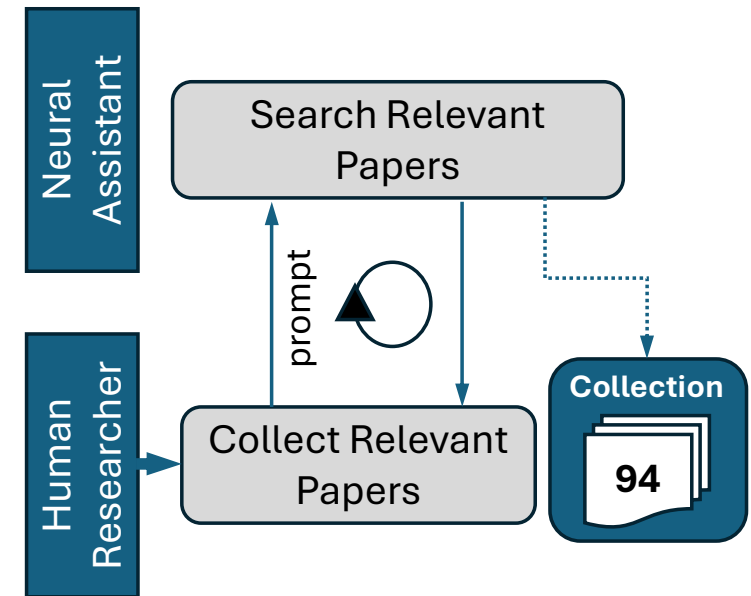
48 papers added to the collection



Domain: Abstraction and Reasoning Challenge

Papers Collected 5/5

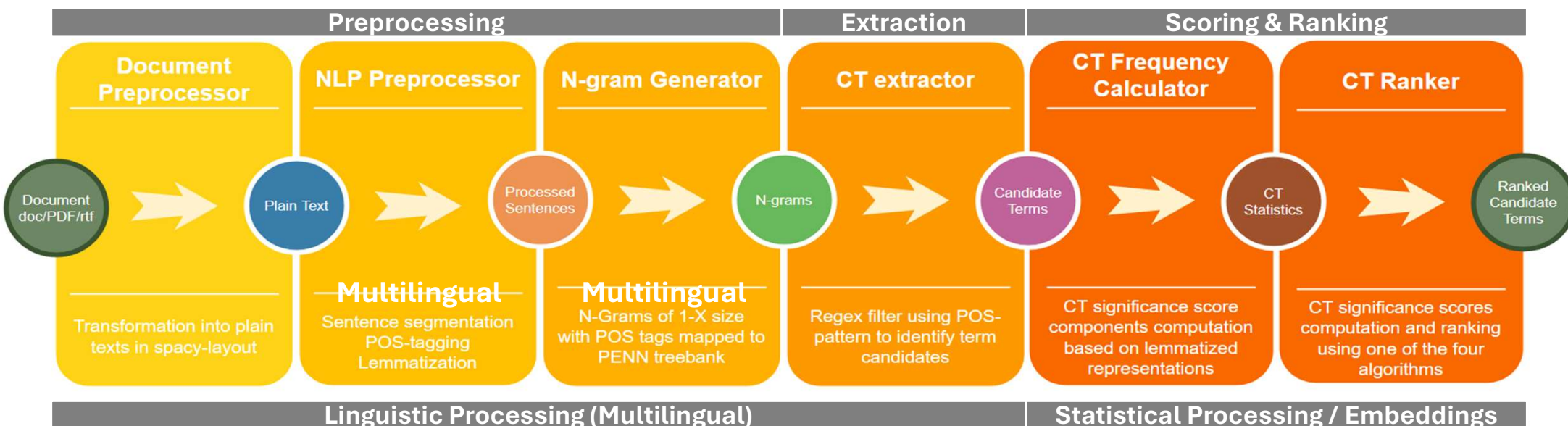
- One of our PhD students
 - Collected the initial set of pubs – manually: ~50 papers
 - Sources: Connected Papers, arXiv, Google Scholar, Kaggle, etc.
 - Prompted ChatGPT 4o – relevance check
 - Prompted ChatGPT 4o – completeness
 - Queried SCOPUS: ~60 more papers
 - Created collection catalogue
 - Took citation frequencies from G-Scholar
 - Downloaded full texts in PDF
 - Some not open-access



Total: **94** full texts

Extractomat*: Term Recognition

Rule-based (L+S) + nested terms (C-Value) + Sentence Embeddings (Rerank) + Multilingual (En, De, Uk + Fr, Nl)



* <https://github.com/lang-uk/extractomat/>

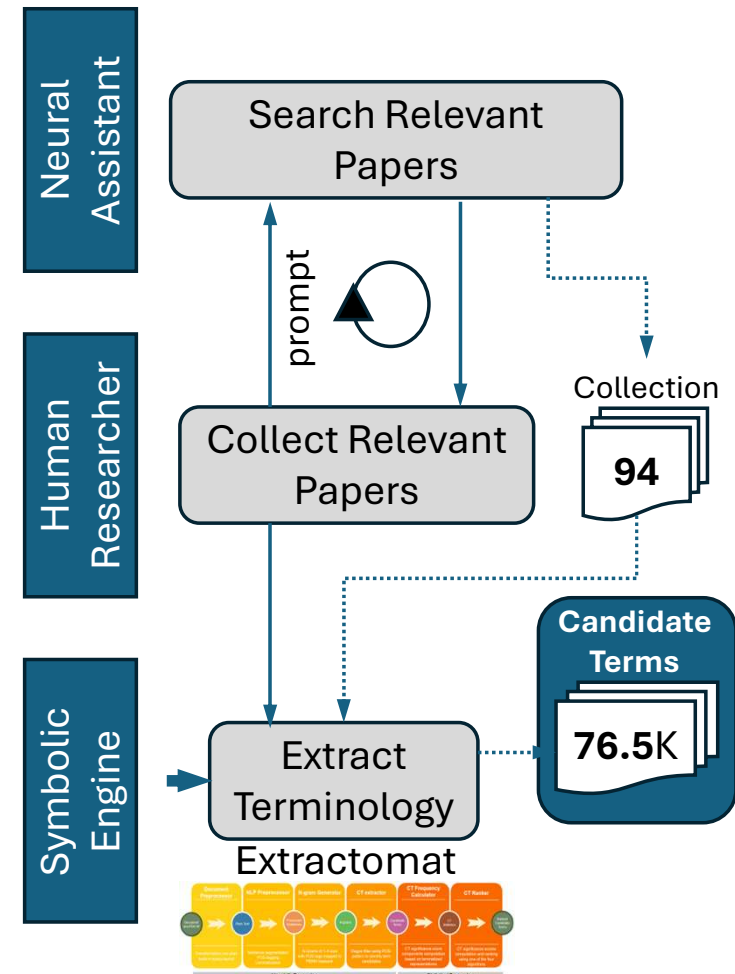
* Chaplynskyi, D., Wittenborg, T., Kosa, V., Rabby, G., Ignatenko, O., Auer, S., and Ermolayev, V.: Building Multilingual Terminological Bridges between Language-Specific Knowledge Silos. CCIS, Vol. 2359, pp. 219 - 232, Springer Science and Business Media Deutschland GmbH (2025) – in press

Domain: Abstraction and Reasoning Challenge

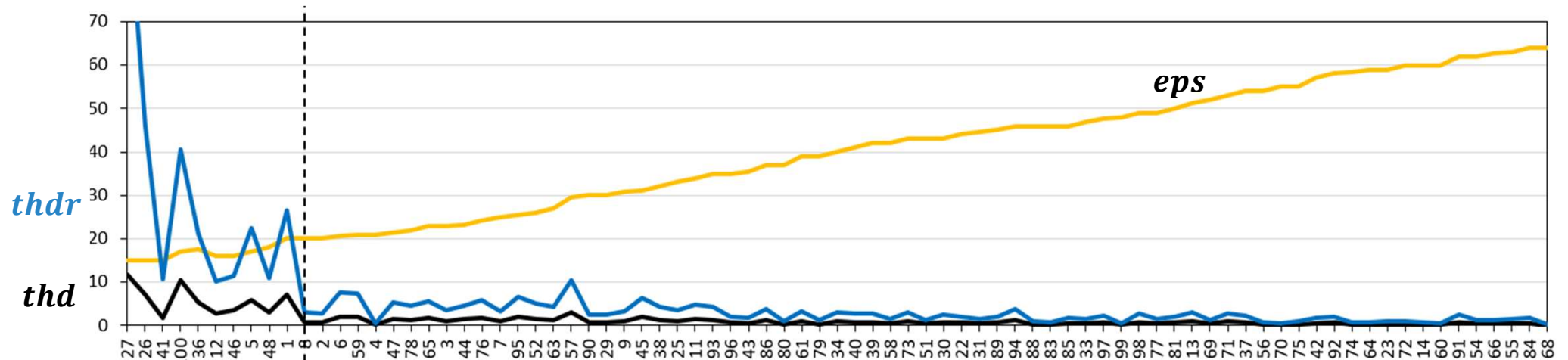
- For every paper in the collection
 - Preprocessing:
 - PDF -> plain text
 - Linguistic Pipeline
 - Statistical Pipeline
 - Scalable C-Value (MPCV)
 - DL-based refinement
 - Reranking
 - n-sentence embeddings
 - Total Candidates: **76.5K**



task	344
intelligence	250
system	248
skill	225
generalization	170
test	151
program	144
t	144
prior	117
evaluation	115
ability	113
human	99
skill program	86.76
knowledge	86
intelligent system	85.59
situation	84
training	75
set	70
?	67
generalization difficulty	62.55
...	



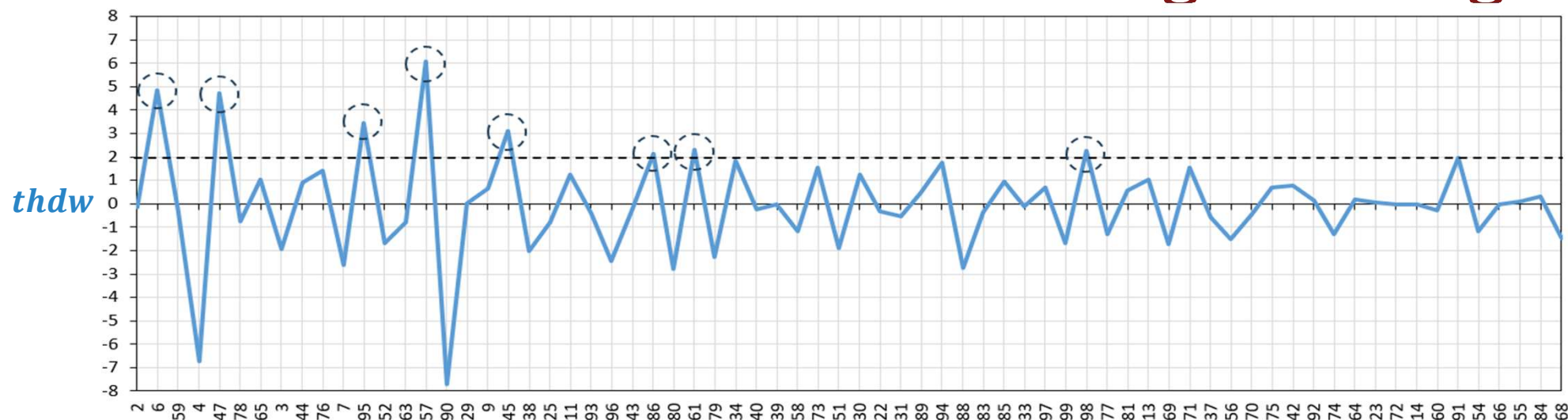
Domain: Abstraction and Reasoning Challenge



Terminology saturation analysis

- Papers ordered by descending citation frequency
- Dataset (D_i) increment – one paper
- Core Sub-Collection: 11 out of 94 papers
 - Carry all the mainstream terms
 - Represent the core of community knowledge

Domain: Abstraction and Reasoning Challenge



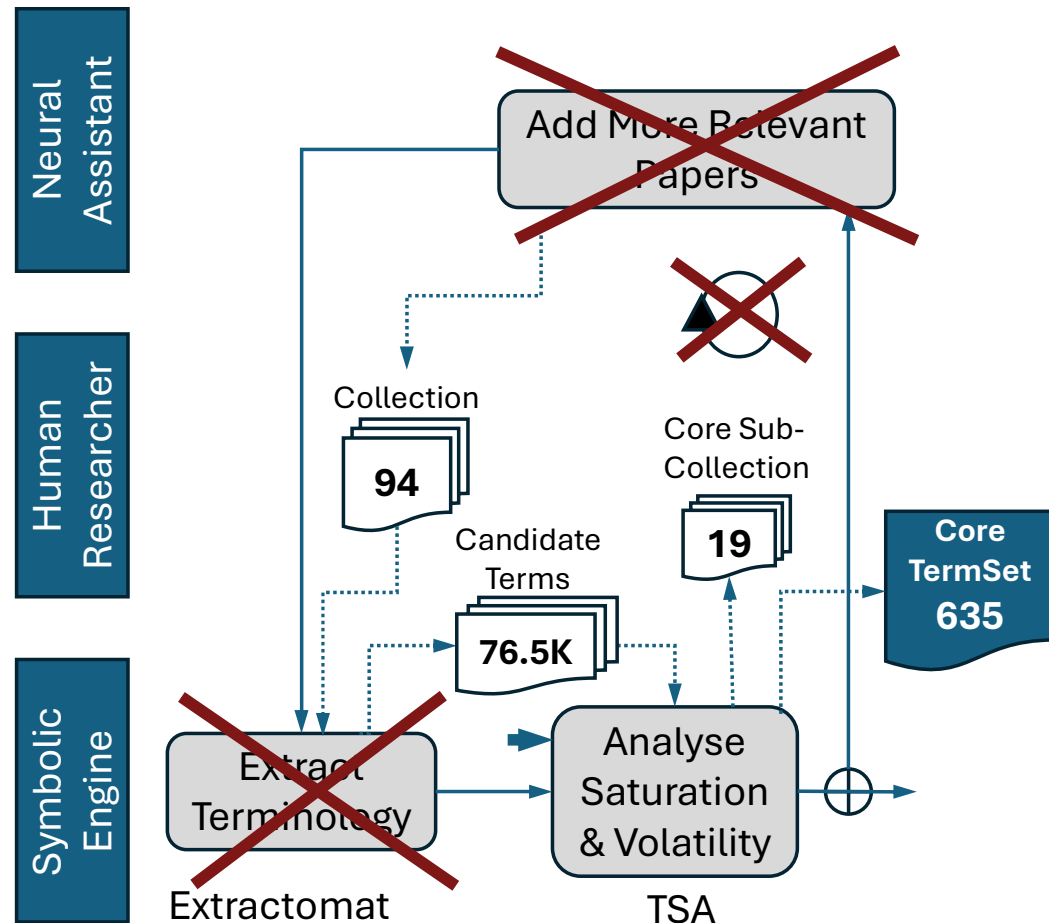
Terminology volatility analysis

- Major terminology contributors identified beyond the Core Sub-Collection
 - 8 more papers
 - Carry all the significant contributions beyond the mainstream
 - Represent domain/topic evolution trends

Domain: Abstraction and Reasoning Challenge

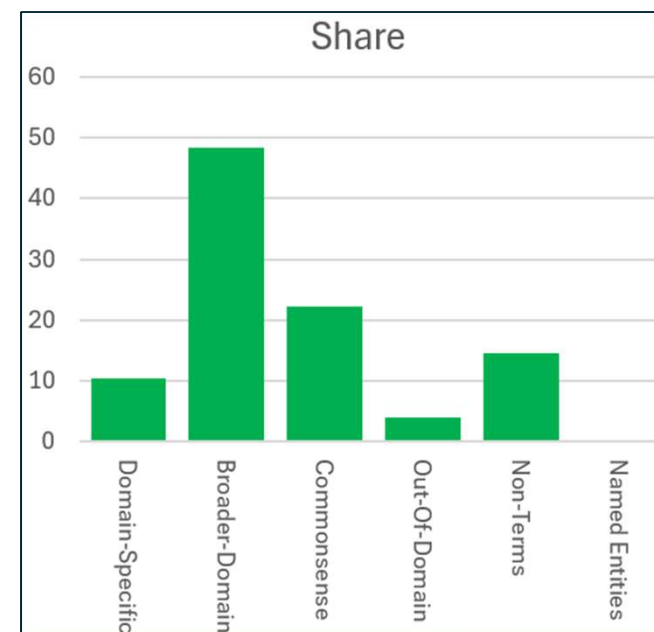
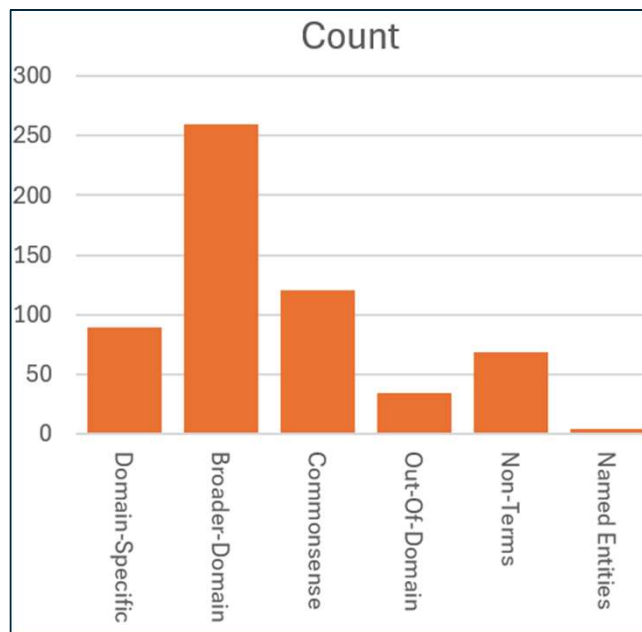
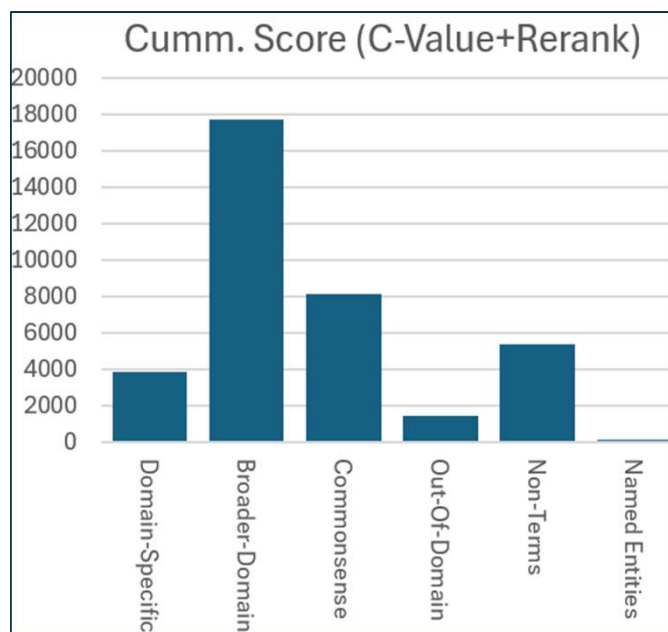
Saturation+Volatility:

- Saturation after the 1st iteration
 - **19** papers to read, out of 94, 20%
- **Completeness:** Contains
 - Mainstream: terminologically saturated
 - Most innovative beyond
- **Explainability:** The result could be explained by labelling texts using the Core TermSet



Domain: Abstraction and Reasoning Challenge

Recognition by Extractomat. Categorisation by a human expert



	Domain-Specific	Broader-Domain	Commonsense	Out-Of-Domain	Non-Terms	Named Entities	Total RST
Cumm. Score	4 795.56	12 928.72	18 653.21	2 114.57	6 381.05	216.89	45 090
Count	89	206	220	37	79	4	635
Score Share (%)	10.64	28.67	41.37	4.05	14.66	0.48	~100.00

Major Terminology Contributing Papers for ARC

Terminological Core Sub-Collection (TCSC)					Major Terminology Contributors Beyond TCSC					
Cat #	Year of Publication	Cit. / Year	Reference	Words	Cat #	Year of Publication	Cit. / Year	thdv	Words	Reference
27	2022	864,00	[1]	22 285	57	2025	20.00	6.0768	15 851	[12]
26	2022	730,00	[2]	19 846	6	2019	94.57	4.8581	6 511	[13]
41	2021	433.00	[3]	6 672	47	2023	82.00	4.6994	10 760	[14]
100	2024	386,50	[4]	21 286	95	2024	41.50	3.4612	8 207	[15]
36	2023	386.00	[5]	16 043	45	2019	11.14	3.1017	17 951	[16]
12	2017	308,67	[6]	8 343	98	2024	2.00	2.2409	9 096	[17]
46	2024	285.00	[7]	10 260	61	2025	5.00	2.3032	17 201	[18]
5	2021	241,66	[8]	8 685	86	2024	6.00	2.141	13 496	[19]
48	2022	201.00	[9]	10 560						
1	2019	126.00	[10]	26 338						
8	2019	122,57	[11]	5 469						

Organization of Our Work

- To join the task force, please fill-out the form

at: <https://forms.gle/e6VnNpGHPjLtaEcw8>

Commitment

What and When

- You will then receive a block of text to annotate
 - Blocks of text are compiled from the ARC papers
 - Every block of text will contain **10-12 K words**
 - Which is ~ the size of **one paper**, on average
- The task will be to annotate the text by **manually extracting terms in categories**
 - As described in the subsequent slides
- The duration of work will be 1 month
 - Hard deadline: **August 5**

Reward

- A successful accomplishment of the task (report submitted) will be rewarded with **2 ECTS**
 - An additional certificate will be provided

Annotating Papers: The Report

A	B	C	D	E	F	G	H
	Sentence	Term					
		Domain-Specific	Evidence for DST	Broader-Domain	Commonsense	Out-of-Domain	Named Entity
	Paper Title: On the Measure of Intelligence						
	Paper Authors: François Chollet						
	Full Citation: Chollet, François. On the Measure of Intelligence. arXiv:1911.01547, arXiv, 25 Nov. 2019. arXiv.org, https://doi.org/10.48550/arXiv.1911.01547						
	To make deliberate progress towards more intelligent and more human-like artificial systems, we need to be following an appropriate feedback signal.			intelligent artificial system	progress		
				human-like artificial system	deliberate progress		
				artificial system	system		
					signal		
					feedback signal		
	We need to be able to define and evaluate intelligence in a way that enables comparisons between two systems, as well as comparisons with humans.				intelligence		
					way		
					comparison		
					system		
					comparison		
	Over the past hundred years, there has been an abundance of attempts to define and measure intelligence, across both the fields of psychology and AI.				human		
					attempt		Psychology
					intelligence		AI
					field		year hundred

Teamwork

- You are **encouraged to team** for
 - Developing code to instrument routine operations:
 - Sentence chunking
 - Part-of-speech tagging (if deemed helpful)
 - Anything else **except terms recognition**
- Some of you may receive overlapping texts to annotate
 - You are **NOT ALLOWED** to share annotations with each other
 - Inter-annotator agreement will be finally measured by us

The Use of AI

- You are allowed to use any LLM-based chatbot, API, or agent to:
 - Verify if a word combination (your term candidate) is either:
 - A term or not a term
 - A Domain-Specific (ARC) Term
 - A Broader-Domain-Specific Term (e.g., AI in general, or Computer Science)
 - A Commonsense Term
 - A Named Entity, being a name of the instance of a concept
 - Not a Term
- You are **NOT allowed** to use AI for **any other purpose** in this task
 - Specifically, you are **FORBIDDEN to use AI to recognize term candidates**

Prompting Gemini for Verification: Term?

“butterfly effect”



You are a knowledge engineer informed about the Abstraction and Reasoning Challenge domain. Please verify if "butterfly ...

No, "butterfly effect" is not a formal, recognized technical term within the specific domain of François Chollet's [Abstraction and Reasoning Challenge \(ARC\)](#).

While it is a foundational concept in chaos theory, its appearance in discussions surrounding ARC is purely informal or analogical.

Why the Analogy is Used Informally

When AI researchers or participants in the ARC Prize mention a "butterfly effect" in the context of LLMs or abstract reasoning, they are usually referring to one of two behaviors:

- **Sensitivity to Prompt Changes (Semantic Brittleness):** In programmatic or LLM-based solvers, an extremely minor tweak to a prompt, a single pixel layout shift, or a tiny variance in the string representation of a grid can lead to a complete collapse in the model's reasoning chain. [Hacker News +1](#)
- **Error Propagation in Generation:** In multi-step programmatic grid transformations (e.g., executing a sequence of generated Python functions to solve an ARC task), a microscopic mistake in an early transformation step completely derails all subsequent grid states. [arXiv](#)

ODT

or



Please verify if it is a broader-domain-specific term in AI

No, "butterfly effect" is not a formal, domain-specific technical term in artificial intelligence.

Instead, it is an imported analogy from mathematics and chaos theory. AI researchers use it informally to describe how tiny changes in a system's inputs or parameters can trigger massive, unpredictable shifts in output. [arXiv +3](#)

While several researchers increasingly invoke the analogy across specific mathematical and systemic phenomena:

1. Algorithmic Unfairness

In AI ethics, a [peer-reviewed paper by Emilio Ferrara](#) formalized the analogy as the "Butterfly Effect in AI Systems". It describes how microscopic biases or skewed data inputs during early training, or minute distribution shifts between training and deployment, propagate through deep neural networks to produce massive, systemic unfairness. [arXiv](#)

2. Sensitivity in Deep Learning Optimization

During the training of deep neural networks, minor changes to initial weight initializations or tiny perturbations near **saddle points** (flat regions in the optimization landscape) can cause gradient descent to take radically different paths. This leads to vastly different final models, showcasing the chaotic nature of deep learning

NT

Prompting Gemini for Verification: DST

DST

Please verify if "core knowledge prior" is a domain-specific term in ARC. If yes, please provide the reference and citation that denotes this term.

^

21:06

Yes, "Core Knowledge priors" is a domain-specific term in ARC. It refers to the foundational, innate human cognitive abilities that the benchmark assumes an intelligence agent must have to solve the puzzles without relying on massive datasets.  arXiv +3

Reference and Citation

The term was coined and formally defined by François Chollet in his seminal 2019 whitepaper that introduced the ARC benchmark.  arXiv +1

- Document Title: [On the Measure of Intelligence](#)
- Author: François Chollet (Google, Inc.)
- Publication Date: November 25, 2019

“core knowledge prior”

! Please provide the **evidence** for BST in your report:

- The citation of the definition
- The reference to the respective paper

Prompting Gemini for Verification: BDST

“intelligent artificial system”

BDST

You are a knowledge engineer informed about the Abstraction and Reasoning Challenge (ARC) domain

Please verify if "intelligent artificial system" is a domain-specific term in ARC



No, **"intelligent artificial system"** is not a domain-specific term in the Abstraction and Reasoning Challenge (ARC) domain.

While ARC focuses on artificial general intelligence (AGI) and evaluating how machines learn new skills, the phrase "intelligent artificial system" is just a general, descriptive arrangement of words used across the broader field of computer science. It does not hold a specific, technical definition inside the ARC framework. 

Prompting Gemini for Verification: NE

“algorithmic information theory”



Please verify if "Algorithmic Information Theory" is a named entity in ARC

Yes, "Algorithmic Information Theory" (AIT) is a fundamental named entity and formal concept in the ARC domain. arXiv

Unlike the phrase "intelligent artificial system," which is just a generic description, Algorithmic Information Theory is a specific, well-established mathematical field that François Chollet directly uses as the foundational backbone for ARC. HTM Forum +3

In his seminal paper, [On the Measure of Intelligence](#), Chollet relies on AIT to build his new mathematical definition of general intelligence. arXiv

How Algorithmic Information Theory (AIT) Functions as a Named Entity in ARC

- **Mathematical Definition of Intelligence:** Chollet uses AIT concepts (like Kolmogorov complexity and minimum description length) to formally define intelligence as **skill-acquisition efficiency**. Substack · AI Changes Everything +1

I look forward to answering your questions

Many thanks to the colleagues who contributed to this work:



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